



 **MLT__OPPE__Mock.ipynb**
File

This is a mock file for my upcoming Machine Learning workshop test. The types of questions in this file reflect what will appear in my online ML exam. Please analyze all the questions and answers in the mock file thoroughly to identify key topics, concepts, and patterns. Based on this analysis, generate 50 high-quality practice questions and answers that closely match the style, difficulty, and scope of the original mock. Cover all major ML topics reflected in the file, including theory, code-based reasoning, model evaluation, preprocessing, and feature selection. First list all 50 questions clearly, then provide detailed answers at the end. Ensure the questions are diverse, exam-focused, and help me score full marks.
Exam is on Numpy and google colab
generate questions for numpy

Thanks for sharing the mock file. Before I generate 50 high-quality practice questions, could you confirm if you'd like them to focus only on NumPy and Google Colab (as mentioned at the end), or should I include other Machine Learning topics present in the mock file as well?



include other Machine Learning topics present in the mock file as well

Great. I'll review the mock test file you provided to extract key topics, concepts, and question formats related to NumPy, Google Colab, and other Machine Learning areas covered. Based on this, I'll generate 50 diverse, high-quality practice questions that reflect the style, difficulty, and scope of the exam. I'll let you know once they're ready.



Generate 50 Practice Questions Based on Machine Learning Mock Exam

Practice Questions

- Outer Product Index:** Let $x = \text{np.array}([1, 2, \dots, 10])$ and $y = \text{np.array}([10, 11, \dots, 21])$. Compute the outer product matrix $M = x[:, \text{None}] * y[\text{None}, :]$. What is the entry $M[8, 9]$ (using 1-indexing)?
- Matrix Multiplication Elements:** If matrix A has shape $(120, 80)$ and B has shape $(80, 25)$, then $M = A @ B$ has shape $(120, 25)$. How many total elements does M contain?
- Euclidean Distance:** Compute the Euclidean distance between the points $(1, 2, 3)$ and $(4, 6, 8)$. Give your answer to two decimal places.
- Z-score Normalization:** Given the data array $[2, 4, 6, 8]$, compute the standardized z-score of the value 8 (subtract the mean and divide by the standard deviation). Provide the answer to

three decimal places.

5. **Precision & Recall:** In a binary classification task, suppose True Positives (TP) = 40, False Positives (FP) = 10, False Negatives (FN) = 20. Compute **precision** and **recall**. (Round each to three decimal places.)
6. **Classification Accuracy:** A classifier makes predictions on 500 samples. Given an array of true labels y (± 1) and predicted labels $\hat{y}$ (± 1), accuracy is defined as the fraction of correct predictions. If 252 of 500 predictions are correct, what is the classification accuracy (to three decimals)?
7. **Linear Regression Weights:** A linear regression model with bias term uses a “dummy” feature of all ones. If the original data has 5 features, how many weights (parameters) does the model learn (including the bias)?
8. **Weight Norm:** A linear regression solution yields weight vector $w^* = [0.5, -1.2, 2.3, 0.0, 0.8, -0.6]$. Compute the Euclidean norm $\|w^*\|$ of this weight vector (two decimals).
9. **Training RMSE:** Given true target values y_{train} and predictions $\hat{y}_{\text{train}}$ on the training set, the root mean squared error (RMSE) is $\sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$. Suppose $\sum (y_i - \hat{y}_i)^2 = 25$ for $n = 100$ training points. What is the training RMSE (two decimals)?
10. **Test RMSE:** On a test set of $m = 50$ points, suppose $\sum (y_i - \hat{y}_i)^2 = 12.25$. What is the test RMSE (two decimals)?
11. **Ridge Regression (Train):** In ridge regression with weight-decay λ , suppose you test λ values $[0.01, 0.1, 1, 10, 100]$ and find the smallest training RMSE occurs at $\lambda = 0.01$. Which λ gives the lowest training error?
12. **Ridge Regression (Test):** Using the same λ candidates, suppose the smallest test RMSE is achieved at $\lambda = 1$. Which λ gives the lowest test error?
13. **Concatenation Shape:** Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}$. Form $H = \text{np.hstack}((A, B))$. How many columns does H have?
14. **Count Zeros:** Let $C = \begin{pmatrix} 0 & -1 & 5 \\ 2 & 0 & -3 \\ 0 & 1 & 0 \end{pmatrix}$. How many entries of C are zero?
15. **Min-Max Normalization:** Suppose the feature range is from 10 to 20, linearly scaled to $[0, 1]$. What is the normalized value of $x = 15$? (Use $(x - 10)/(20 - 10)$ formula.)
16. **One-Hot Encoding Count:** A dataset has one numeric feature and two categorical features: “Color” with 3 categories (Red, Green, Blue) and “Size” with 4 categories (S, M, L, XL). Using one-hot encoding **without dropping any category**, how many total features are there? (Count the numeric and all binary columns.)
17. **K-Fold CV Split:** If you have 100 samples and perform 5-fold cross-validation (equal-sized folds), how many samples are in each fold?
18. **np.arange Last Value:** What is the last value of `np.arange(0, 30, 7)`?

19. **Fancy Indexing Sum:** Given array $A = \text{np.array}([5, 10, 15, 20])$, compute $A[[0,2]]$ and report its sum.
20. **Broadcast Addition:** Let $A = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ (shape 3×1) and $B = [10, 20]$ (length-2 row). After computing $M = A + B$, what is the entry $M[3,2]$ (1-indexed)?
21. **Transpose Element:** Let $M = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$. What is $M^T[1,3]$ (1-indexed)?
22. **Unique Count:** For array $D = \text{np.array}([1,2,2,3,3,3])$, how many unique elements does it contain?
23. **F1 Score:** If a classifier has precision 0.80 and recall 0.75, what is the F1 score? (Round to two decimals.)
24. **Log Loss (Cross-Entropy):** For a single sample with true label $y = 1$ and predicted probability $p = 0.8$, the logistic log loss is $-[y \log p + (1 - y) \log(1 - p)]$. Compute this log loss (two decimals).
25. **R² (Coefficient of Determination):** Let true values be $y = [1, 2, 3]$ and predictions $\hat{y} = [2, 2, 2]$. Compute $R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$ (two decimals).
26. **Pearson Correlation:** Compute the Pearson correlation between $X = [1, 2, 3]$ and $Y = [2, 4, 6]$. (Two decimals.)
27. **Argmax Index:** Given $a = \text{np.array}([5, 7, 2, 10])$, what index does $\text{np.argmax}(a)$ return? (0-indexed.)
28. **Specificity:** In a confusion matrix, specificity (true negative rate) is $TN / (TN + FP)$. Given $TN=80$ and $FP=10$, what is specificity? (Three decimals.)
29. **Balanced Accuracy:** Balanced accuracy is defined as the average of sensitivity and specificity. Using sensitivity = 0.818 and specificity = 0.889 (from previous examples), what is the balanced accuracy? (Three decimals.)
30. **Frobenius Norm:** The Frobenius norm of a matrix is the square root of the sum of squares of all entries. Compute the Frobenius norm of $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. (Round to three decimals.)
31. **Trace of a Matrix:** Compute the trace (sum of diagonal entries) of $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.
32. **Determinant (2x2):** Compute the determinant of $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. (Use formula $ad - bc$.)
33. **Sum of Eigenvalues:** A 2×2 diagonal matrix has eigenvalues 2 and 3. What is the sum of its eigenvalues?
34. **Product of Eigenvalues:** For the same matrix (eigenvalues 2 and 3), what is the product of its eigenvalues?

35. **Inverse Element:** Let $N = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$. Its inverse is $\frac{1}{\det(N)} \begin{pmatrix} 4 & -3 \\ -1 & 2 \end{pmatrix}$. What is the (1, 1) entry of N^{-1} ? (Two decimals.)
36. **Matrix Rank:** What is the rank of $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$?
37. **Vector Norms:** For vector $v = [1, -3, 4]$, compute the L1 norm ($\|v\|_1$) and L2 norm ($\|v\|_2$). (Give each to one decimal.)
38. **Sigmoid Function:** Compute the logistic sigmoid of 0.8, i.e. $\sigma(0.8) = 1/(1 + e^{-0.8})$. (Three decimals.)
39. **Standard Deviation:** Compute the population standard deviation of $[1, 2, 3]$. (Two decimals.)
40. **Dot Product:** Compute the dot product of $[1, 2, 3]$ and $[4, 5, 6]$.
41. **Argmin Index:** Given `b = np.array([4,1,9])`, what index does `np.argmin(b)` return? (0-indexed.)
42. **Reshape Index:** Create `E = np.arange(1,13).reshape(3,4)`. What is the entry `E[2,3]` (1-indexed)?
43. **Vector L2 Norm:** Compute the L2 norm of the vector $[3, 4]$.
44. **Range:** Given the list $[2, -1, 5]$, compute the range (max minus min).
45. **Sum of Elements:** Compute the sum of the array $[1, 2, 3, 4]$.
46. **Arithmetic Mean:** Compute the mean of the array $[1, 2, 3, 4]$. (One decimal.)
47. **Variance:** Compute the population variance of $[1, 2, 3, 4]$. (Use $\sigma^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$. Two decimals.)
48. **Cosine Similarity:** Compute the cosine similarity between vectors $[1, 0]$ and $[0, 1]$. (Two decimals.)
49. **Element-wise Maximum:** Given $[2, -1, 5]$, what is the maximum element?
50. **Sum of Dimensions:** A matrix has shape (7,8). What is the sum of its dimensions (7 + 8)?

Detailed Answers

1. **Outer product:** The outer product $M_{ij} = x_i \times y_j$. Here $x_8 = 8$ and $y_9 = 18$ (since y runs from 10 to 21). So $M[8, 9] = 8 \times 18 = 144$. This follows $M[i, j] = x_i y_j$ numpy.org.
2. **Matrix elements:** The product $M = A @ B$ has shape (120,25), so it contains $120 \times 25 = 3000$ elements.
3. **Euclidean distance:** Distance = $\sqrt{(4-1)^2 + (6-2)^2 + (8-3)^2} = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{9 + 16 + 25} = \sqrt{50} = 7.071$ (to two decimals). Euclidean distance is defined by the Pythagorean formula en.wikipedia.org.
4. **Z-score:** The mean is $(2 + 4 + 6 + 8)/4 = 5$, std = $\sqrt{[(2-5)^2 + (4-5)^2 + (6-5)^2 + (8-5)^2]/4} = \sqrt{(9 + 1 + 1 + 9)/4} = \sqrt{20/4} = \sqrt{5} = 2.236$. The z-score for 8 is $(8-5)/2.236 = 3/2.236 = 1.342$ (three decimals). Standard score $z = (x - \mu)/\sigma$ en.wikipedia.org.
5. **Precision & Recall:** Precision = $\frac{TP}{TP+FP} = 40/50 = 0.800$. Recall = $\frac{TP}{TP+FN} = 40/60 = 0.667$. (To three decimals: 0.800 and 0.667.) These follow Precision = $TP/(TP + FP)$, Recall = $TP/(TP + FN)$ [developers.google.com](https://developers.google.com/developers.google.com).

6. **Accuracy:** Accuracy = (# correct) / (total) = 252/500 = 0.504. Equivalently, accuracy is the proportion of correct predictions evidentlyai.com. So 0.504 (three decimals).
7. **Weights count:** With 5 original features plus 1 dummy (bias) feature, the model learns 6 weights in total.
8. **Weight norm:** $\|w^*\| = \sqrt{0.5^2 + (-1.2)^2 + 2.3^2 + 0^2 + 0.8^2 + (-0.6)^2} = \sqrt{0.25 + 1.44 + 5.29 + 0 + 0.64 + 0.36} = \sqrt{7.98} = 2.827$ (2.83).
9. **Train RMSE:** RMSE = $\sqrt{25/100} = \sqrt{0.25} = 0.50$.
10. **Test RMSE:** RMSE = $\sqrt{12.25/50} = \sqrt{0.245} = 0.495$ (0.50).
11. **Best λ (train):** The smallest training error occurs at $\lambda = 0.01$.
12. **Best λ (test):** The smallest test error occurs at $\lambda = 1$.
13. **Hstack columns:** $H = [A \mid B]$ has 2 rows and $2 + 2 = 4$ columns. So the number of columns is 4.
14. **Count zeros:** The zeros in C are at entries (1,1), (2,2), (3,1), (3,3) - a total of 4 zero entries.
15. **Min-max norm:** $(15 - 10)/(20 - 10) = 5/10 = 0.5$.
16. **One-hot count:** We get 3 binary features for "Color" and 4 for "Size", plus 1 numeric feature = $3 + 4 + 1 = 8$ features in total [geeksforgeeks.org](https://www.geeksforgeeks.org).
17. **K-fold split:** 100 samples / 5 folds = 20 samples per fold (equal-sized) scikit-learn.org.
18. **np.arange(0,30,7):** This produces [0, 7, 14, 21, 28]. The last value is 28.
19. **Fancy indexing sum:** $A[[0,2]] = [5,15]$. Sum = $5 + 15 = 20$.
20. **Broadcast sum:** $A + B$ yields a 3×2 matrix. Row 3, Column 2 (1-index) corresponds to $3 + 20 = 23$. So $M[3,2] = 23$. (See broadcasting example [jakevdp.github.io](https://github.com/jakevdp).)
21. **Transpose element:** $M^T = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$. The entry (1,3) is 5.
22. **Unique count:** The unique values in [1, 2, 2, 3, 3, 3] are {1, 2, 3}. Count = 3.
23. **F1 score:** $F1 = 2 \times \frac{0.80 \times 0.75}{0.80 + 0.75} = 2 * (0.600/1.55) = 0.774$. Rounded = **0.77** (two decimals).
Formula $F1 = 2 \frac{Precision \times Recall}{Precision + Recall}$ developers.google.com.
24. **Log loss:** $-[1 \cdot \ln(0.8) + 0 \cdot \ln(0.2)] = -\ln(0.8) = 0.223$. (Three decimals: **0.223**.)
25. **R^2 :** $\bar{y} = 2$. $SSE = (1 - 2)^2 + (2 - 2)^2 + (3 - 2)^2 = 1 + 0 + 1 = 2$. $SST = (1 - 2)^2 + (2 - 2)^2 + (3 - 2)^2 = 1 + 0 + 1 = 2$. So $R^2 = 1 - 2/2 = 0.00$. en.wikipedia.org
26. **Correlation:** $\text{corr}(X, Y) = 1.00$ (perfect linear scaling). [Since $Y = 2X$, the Pearson $r = +1$ en.wikipedia.org.]
27. **Argmax:** `np.argmax([5,7,2,10])` returns index **3** (the max 10 is at position 3, 0-indexed) numpy.org.
28. **Specificity:** $= TN/(TN + FP) = 80/(80 + 10) = 80/90 = 0.8889$. (Three decimals: **0.889**.) poojapawani.medium.com
29. **Balanced accuracy:** $(0.818 + 0.889)/2 = 0.8535$. (Three decimals: **0.854**.) Balanced accuracy is the average of sensitivity and specificity scikit-learn.org.

30. **Frobenius norm:** $\sqrt{1^2 + 2^2 + 3^2 + 4^2} = \sqrt{1 + 4 + 9 + 16} = \sqrt{30} = 5.477$ (5.477 to three decimals)help.perforce.com.
31. **Trace:** $1 + 4 = 5$. (Trace is sum of diagonalstatlect.com.)
32. **Determinant:** $1 * 4 - 2 * 3 = 4 - 6 = -2$ en.wikipedia.org.
33. **Sum of eigenvalues:** For a diagonal matrix with eigenvalues 2 and 3, the sum is $2 + 3 = 5$. (This equals the trace.)
34. **Product of eigenvalues:** $2 \times 3 = 6$. (This equals the determinant.)
35. **Inverse element:** $\det(N) = 5$. The (1,1) entry of N^{-1} is $4/5 = 0.80$.
36. **Rank:** The rows (or columns) of $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ are linearly dependent, so rank = 1en.wikipedia.org.
37. **Vector norms:** $\|v\|_1 = |1| + |-3| + |4| = 1 + 3 + 4 = 8.0$. $\|v\|_2 = \sqrt{1^2 + (-3)^2 + 4^2} = \sqrt{1 + 9 + 16} = \sqrt{26} = 5.1$.
38. **Sigmoid:** $\sigma(0.8) = 1/(1 + e^{-0.8}) = 0.68997$. (Three decimals: **0.690**.) Using the logistic formulaen.wikipedia.org.
39. **Standard deviation:** For $[1, 2, 3]$, mean = 2. Variance = $(1 + 0 + 1)/3 = 0.667$. Std = $\sqrt{0.667} = 0.816$. (Two decimals: **0.82**.)
40. **Dot product:** $[1, 2, 3] \cdot [4, 5, 6] = 1 * 4 + 2 * 5 + 3 * 6 = 4 + 10 + 18 = 32$ en.wikipedia.org.
41. **Argmin:** `np.argmax([4,1,9])` returns index **1** (the minimum 1 is at position 1, 0-indexed).
42. **Reshape element:** `E = np.arange(1,13).reshape(3,4)` creates

```
lua
```

```
E = [[ 1,  2,  3,  4],
      [ 5,  6,  7,  8],
      [ 9, 10, 11, 12]]
```

The entry at row 2, col 3 (1-indexed) is **7**.

43. **L2 norm [3,4]:** $\sqrt{3^2 + 4^2} = 5.0$.
44. **Range:** max=5, min=-1, range = $5 - (-1) = 6$.
45. **Sum of [1,2,3,4]:** $1 + 2 + 3 + 4 = 10$ numpy.org.
46. **Mean of [1,2,3,4]:** $(1 + 2 + 3 + 4)/4 = 2.5$ numpy.org.
47. **Variance of [1,2,3,4]:** Mean = 2.5, then variance = $[(1.5^2 + 0.5^2 + 0.5^2 + 1.5^2)/4] = (2.25 + 0.25 + 0.25 + 2.25)/4 = 5/4 = 1.25$. (Two decimals: **1.25**.) Using `var =  $\sum (x_i - \bar{x})^2 / n$` numpy.org.
48. **Cosine similarity:** Vectors $[1, 0]$ and $[0, 1]$ are orthogonal. Cosine similarity = 0.
49. **Maximum element [2,-1,5]:** The maximum is **5**.
50. **Sum of dimensions:** For shape (7,8), the sum of dimensions = $7 + 8 = 15$.

Sources: Definitions and formulas used above are standard (e.g. Euclidean distance, z-score, accuracy, precision, etc. from common ML references). Specific formula citations: matrix outer productnumpy.org, accuracyevidentlyai.com, precision/recalldevelopers.google.com

developers.google.com, balanced accuracy scikit-learn.org, specificity poojapawani.medium.com, R^2 en.wikipedia.org, Pearson correlation en.wikipedia.org, dot product en.wikipedia.org, trace statlect.com, determinant en.wikipedia.org, Frobenius norm help.perforce.com, one-hot encoding geeksforgeeks.org, and arithmetic mean numpy.org.



Sources